

Investigations into the causes of residue variability on carrots in the UK

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A pilot study was established to determine whether the preferential flow of water and thus insecticide in carrot beds might be responsible for the high residues of organophosphorus insecticides detected in individual carrot roots grown in the UK. A field site typical of UK carrot growing conditions was selected on a sandy soil with a low organic carbon content. Brilliant blue dye was applied in water to a small number of field plots located in the carrot beds to trace water movement through the bed and not to simulate insecticide application or irrigation. The plots were excavated following sufficient time for infiltration and drainage. Horizontal and vertical cross-sections through the soil profile were cut and descriptions of the dye presence in relation to soil features and the position of the carrot roots were made. Dye tracing and soil analyses showed there was a clear mechanical cultivation effect which generated a preferential movement of dye and water within the bed. The subsequent growth of carrots also created additional pathways of preferential movement due to stem-flow or canopy drip. A second study which increased replication of samples and allowed analysis of triazophos and chlorfenvinphos residues in the carrots could not identify any single factor which was conclusively responsible for initiating high residues in individual roots.

Keywords: organophosphorus insecticides, residues, carrots, preferential flow, dye tracing

Introduction

Analyses of organophosphorus (OP) and carbamate residues in individual carrot roots in the UK revealed a higher than expected degree of variability between individual roots. The residue levels have, in some cases, exceeded Maximum Residue Limits (MRLs) and given rise to potential concern for human health. A number of research studies have taken place over the last few years to investigate the causes and extent of variability in residues and these were concerned with pesticide application technology and machinery, crop and environmental variables. A review for the UK Ministry of Agriculture Fisheries and Food (MAFF) found that agricultural operations had the potential to introduce 3–4-fold variation in residues on individual items whilst crop structure characteristics could cause additional variability (Pesticides Safety Directorate 1994). Environmental factors such as soil organic carbon content, soil pH, soil moisture and soil temperature also contributed to residue variability. The final residue concentration becomes a product of all the agricultural, environmental and crop structure parameters and the sources of variability would tend to combine multiplicatively. It was suggested that extreme residue values might be prevented by reducing the variability in any of the contributory factors. The review concluded that environmental factors are the most difficult to control and are probably the greatest contributor to residue variability.

No previous studies have investigated whether preferential movement of water and soluble pesticide might occur in carrot-growing beds and whether the process might be responsible for the extreme residue values found in individual roots. Previous work (Flury *et al.* 1994, Jarvis 1998, Brown *et al.* 1999) has shown the importance of preferential flow in a range of soil types, including sands, in determining the movement of water and soluble chemicals. Such flow can be responsible for transporting high concentrations of pesticide to discrete locations in soil. This

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paper reports on the initial investigations which were carried out to determine whether preferential flow was a significant process in what is traditionally considered to be a homogeneous growth media (where chromatographic type flows had been anticipated) and then reports on the second phase of investigation where the concentration of triazophos and chlorfenvinphos in individual carrot roots were evaluated in relation to management practices and soil processes.

Overview of UK management practices

Carrots are typically grown in sandy or organic soils in England and Wales (figure 1). A stoneless, free draining, well aerated, loosely packed seed bed is necessary for a good carrot crop. The typical intensive

treatments or cultivations carried out in the UK in accordance with normal agricultural practice of one grower with a loamy sand soil are shown in table 1.

Experimental procedures

A site typical of carrot production was selected in spring 1996 near Newark in Nottinghamshire, UK. The topsoil texture was a loamy sand of the Cuckney series and this was underlain by a sand or loamy sand subsoil. The organic carbon content of the topsoil was approximately 1%. Nine replicate plots were established which were 1.5 m long and covered half of the carrot bed width incorporating two rows of carrots. Brilliant blue dye (Erioglaucine A) was applied in water (4 g l^{-1}) to three plots on three separate occasions to simulate insecticide application and sub-

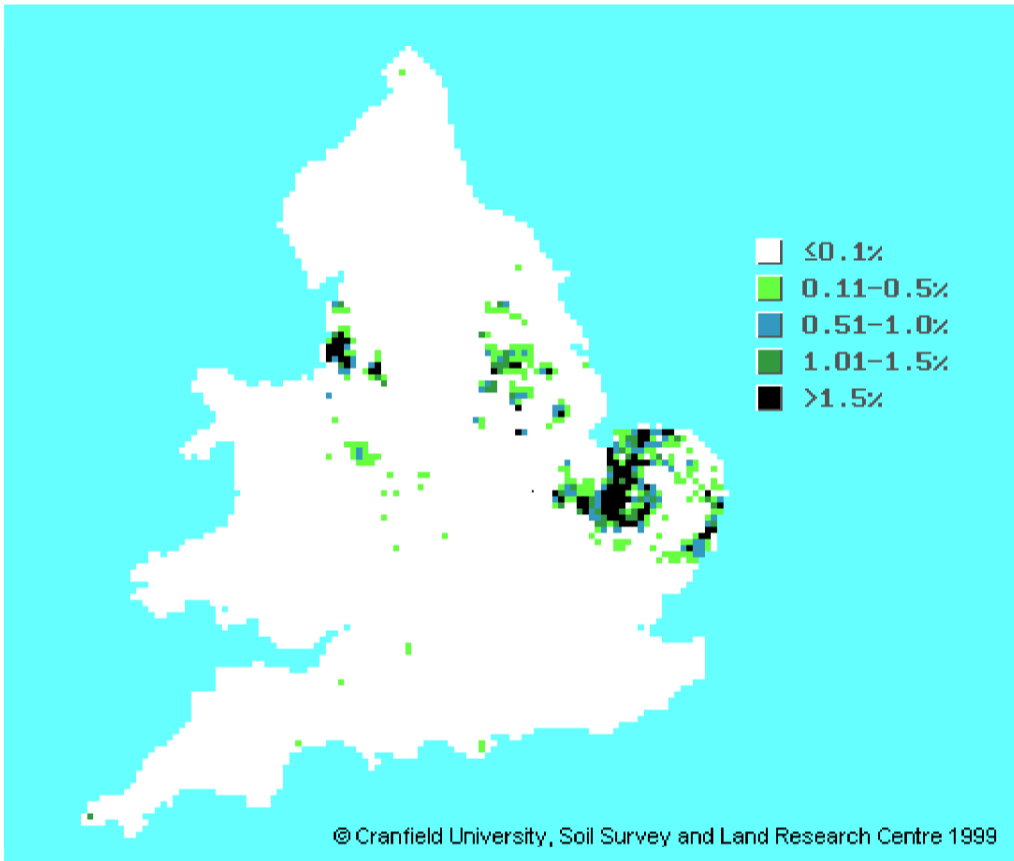


Figure 1. Distribution of carrots on sandy and organic soils in England and Wales.

Table 1. *Treatments and cultivations applied to a typical crop of carrots.*

1	Applied glyphosate herbicide for control of twitch grass in headland areas and also diquat and paraquat for control of weeds and volunteers in the field
2	Dragged soil surface level with a straight tined cultivator (the previous crop was potatoes)
3	Ploughed 20–25 cm deep (end of April)
4	Applied lime to increase pH
5	Ridge formed to aid bed formation and de-stoning
6	De-stoned
7	Bed formed to create three, 72 inch (1.83 m) beds in one pass
8	Drilled beginning of May using a four drill unit with each unit being multi-lined i.e. three lines of seed in each of four rows. (The shoe creates the furrow, the seed coulter delivers the seed and the furrow press replaces and firms the excavated soil)
9	Insecticide was applied as aldicarb granules with the carrot seed within the rows
10	The established seed bed was treated with mancozeb and metalaxyl (systemic and protectant fungicides for cavity spot)
11	Applied metoxuron in July, a contact and residual herbicide to treat volunteer potatoes
12	Applied trace elements manganese, boron, copper and zinc in July
13	Applied cypermethrin to control aphids in July
14	Commenced application of other insecticides as required
15	Applied irrigation, approximately 25 mm every week during the period of soil moisture deficit

sequent transport into the soil by rainfall or irrigation (approximately 10 mm equivalent). The dye was applied immediately after mechanical cultivations were completed in order to evaluate the influence of the seed bed itself on flow pathways and then in July and September when applications of OP insecticides are typically made. On the latter two dates the carrot leaf canopy was present and leaf interception and canopy drip were further factors to consider so the dye was applied to additional plots with either the leaf canopy present or with the canopy removed by cutting off stems at the soil surface. On each occasion the carrot beds were excavated in horizontal and vertical sections at least 24 h after dye applications and distribution of the dye was noted. Undisturbed core samples of soil were also taken from the seed bed to determine dry bulk density. Full details were reported by Fogg and Carter (1997, unpublished).

Observations

Four rows of indentations were observed after the seed beds had been prepared, where the carrot seed and the aldicarb granules were placed by the drill unit. The indentations were created by the furrow shoe which moved soil aside. After seed placement the furrow press firmed the soil back into place. Whilst replication is restricted, the dry bulk densities of the undisturbed soil cores showed the effect of this pressing (table 2).

Table 2. *Dry bulk density in the seed bed*

Soil depth (cm)	Replicate	Bulk density (g cm^{-3})	
		Within drill row	Out of drill row
0–5	A	1.19	1.14
0–5	B	1.18	1.17
5–10	A	1.20	1.12
5–10	B	1.22	1.13

Dye was found to concentrate in the drilled rows and was either absent or less prevalent in between the drilled rows (figure 2). The penetration of the dye was generally homogeneous for the first 2 cm but underneath this layer preferential movement had occurred in places to between 10 and 20 cm depth. The ‘fingers’ of dye measured up to 5 cm in width (figure 3). As carrot growth progressed cracks were observed to develop in the seed bed around individual carrots and between rows despite regular irrigation and maintenance of the soil water status at or near field capacity for most of the time. Dye staining revealed preferential infiltration to 24 cm depth in crack locations and some carrots had a blue surface coloration whilst others did not.

Within each carrot row (comprising three lines of carrots) it was observed that secondary roots of the outside edges of the outside lines were not generally stained blue. Whereas secondary roots adjacent to the middle line were stained (figure 4). Traces of hori-

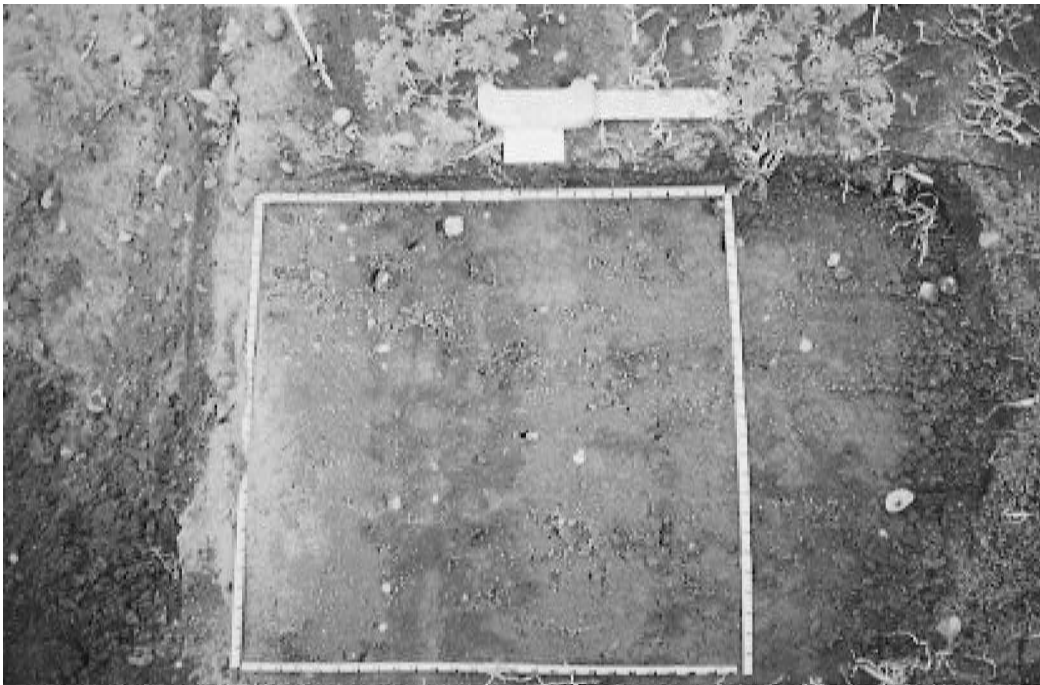


Figure 2. Concentration of dye in the drilled rows at 10 cm depth.



Figure 3. Penetration of dye in the drilled row showing 'fingering'.

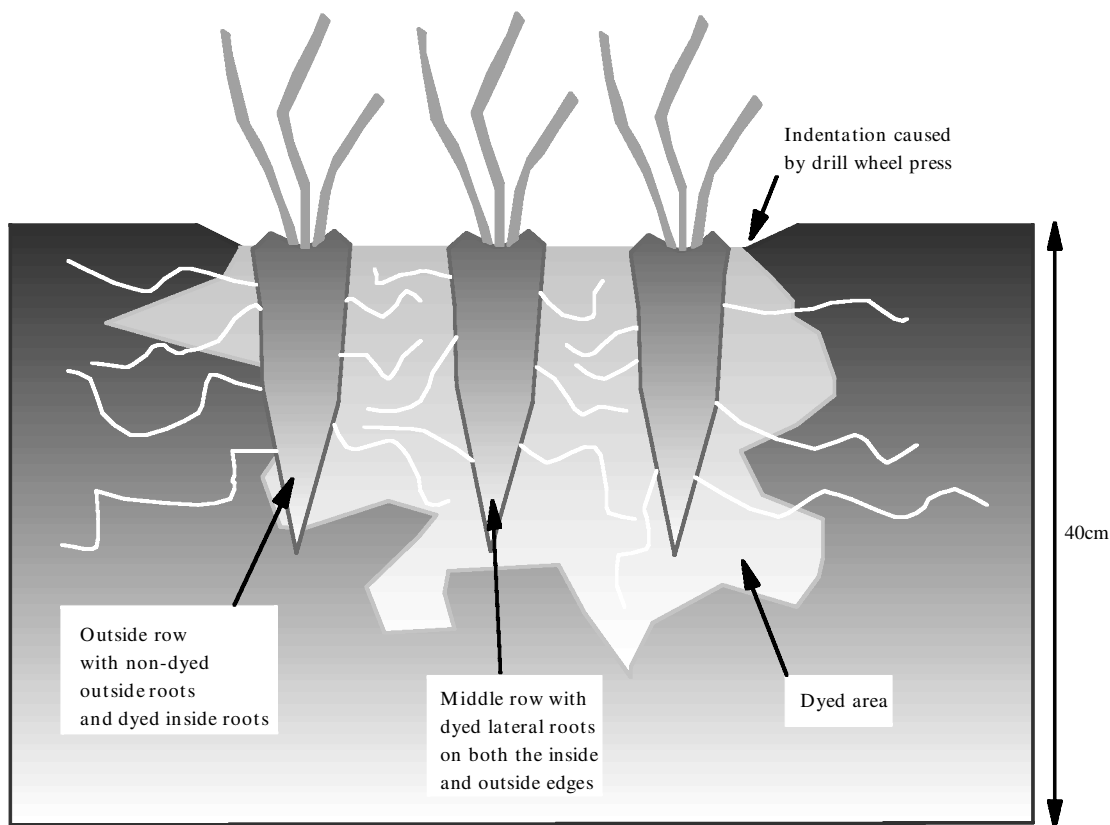


Figure 4. Differential staining of secondary roots in the carrot row.

zontal banding or striping were observed but the effect was less apparent than after bed formation due to carrot growth. On plots where the leaf canopy was retained, dye staining was much more variable than in those where the canopy had been removed.

Discussion

A typical density for this topsoil under cereal production would be $1.4\text{--}1.55\text{ g cm}^{-3}$. The values measured in the carrot seed bed (in and out of rows) were significantly lower ($1.12\text{--}1.22\text{ g cm}^{-3}$) and confirm the very loose packing of the soil. The furrow press slightly increases the soil density and decreases the amount of coarse soil pores. The parts of the soil with a higher proportion of fine pores would therefore have a greater unsaturated hydraulic conductivity and thus more water and pesticide can be preferen-

tially transported. The mechanical cultivation effect on density was identified to be directly responsible for this preferential movement. The carrots, which were drilled in the centre line of a row, had the potential to take up water and dye all around the root. The outside lines of carrots took water from some areas where dye was not prevalent.

The growth of the carrot in the bed may have caused localized 'micro' compaction around the tap root. This slight density change could be responsible for concentrating canopy wash-off and directing stem-flow. The development of cracks was not due to clay shrinkage since these soils do not have sufficient clay content. Cracks are likely to have developed as soil particles pull together with the increased soil water suction as the surface dries.

The combined effects of mechanical cultivation and the influence of the carrot plant itself meant that individual carrots roots could, potentially, be in contact with very concentrated areas of dye whilst surrounding carrot roots are not.

This study was a pilot investigation, was limited in replication and no samples were taken to confirm whether those individual roots which were suspected to contain high residues did so.

Additional studies

A further programme of work was initiated to extend the pilot study to include a greater number of replicates and sites for observations of cultivation practice, dye tracing studies, targeted sampling of individual roots and to enable the control of typical insecticide applications, triazophos and chlorfenvinphos. Residues in selected carrots and some soil samples were then determined.

All sites confirmed the effect of mechanical cultivation on soil density and the associated movement of dye. Soil analyses showed that pesticide residues were higher in the dye stained areas compared with unstained areas. Carrots were selected for analysis on the basis of dye intensity, position in the row, and the degree of exposure of the root crown. Analyses showed residues of triazophos and chlorfenvinphos varying from <0.1 to $>0.5 \text{ mg kg}^{-1}$. Within a plot triazophos residues varied from the lowest to the highest concentration by a factor of up to seven and since carrots were sampled before the required harvest interval in order to maximize the range of con-

centrations many exceeded the MRL of 0.1 mg kg^{-1} (figure 5). Residues of chlorfenvinphos varied from the lowest to the highest concentration by a factor of up to 58.

Analyses showed that there were no statistically significant relationships between residue levels and the extent of dye staining on the surface of carrots or with the extent of exposure of the root crown. There was however some evidence that position in the bed (centre, or outside rows) might account for higher residues, i.e. the highest residues were seen to occur on roots sampled from the outside left row (figure 6) which was in contrast to the hypothesis developed in the pilot study. Beard (1998, unpublished) concluded that despite the increase in replication, insufficient carrots were analysed to provide a statistically significant number of analyses for each treatment, growth stage and site and that any future study should be restricted to a single insecticide and management technique.

Conclusion

Both studies confirmed that there are a number of factors which determine variability of residues in carrot roots. Variability and high concentrations in the OP residues were measured but no conclusive evidence was obtained which identified a particular

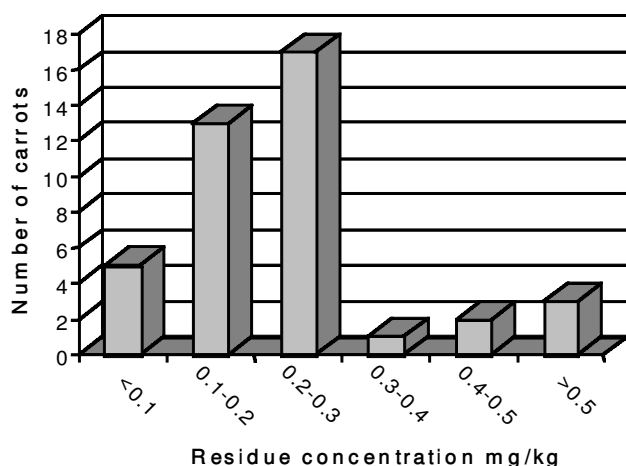


Figure 5. Distribution of triazophos residues (mg kg^{-1}).

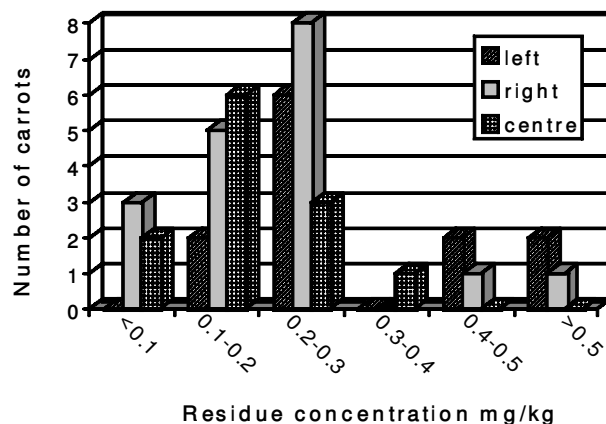


Figure 6. Triazophos residues (mg kg^{-1}) relative to position in row.

cause. Further studies which restrict the number of experimental variables and increase the number of residue analyses are required to provide a more definitive assessment.

Acknowledgements

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